Sukanya Govindaraaj

Technical Business Analyst

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PROFESSIONAL SUMMARY

Technical Product & Business Analyst with 12+ years of end-to-end delivery experience across IT business analysis, enterprise software development, and modern AI/ML frameworks. MBA and PMP-certified with a strong foundational background as a software developer, excelling at bridging the gap between business strategy and complex engineering teams. Skilled in translating high-level business goals into precise epics, user stories, and technical acceptance criteria. Leverages a functional understanding of emerging technologies—including Generative AI, LLMs, and MLOps—to evaluate feasibility, scope intelligent product features, and drive business value across Agile and traditional delivery environments.

EDUCATION & CERTIFICATIONS

Master of Business Administration (MBA) — Simon School of Business, University of Rochester, Rochester, NY (January 2013 – June 2014)

Bachelor of Engineering, Computer Science Engineering — College of Engineering Guindy, Anna University, Chennai, India (August 2005 – June 2009)

PMI Project Management Professional (PMP) - Certified 2023

Great Learning/ University of Texas at Austin — AI/ML Program, 2025- 2026

AWS Certified Cloud Practitioner – Valid from 07/2024 – 07/2027

TECHNICAL SKILLS

Product & Agile Management

PMP, Agile, SCRUM, Kanban, Waterfall, Epics & User Stories, Acceptance Criteria, Requirements Gathering, Stakeholder Management, UAT, JIRA, Azure DevOps, Git/GitHub

AI/ML Frameworks & Methodologies

Generative AI, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Prompt Engineering, Computer Vision, Convolutional Neural Networks (CNNs), Transfer Learning, Predictive Modeling, Classification & Ensemble Methods (Bagging, Boosting, Stacking), Neural Networks

MLOps & Deployment

Docker, Flask API, Streamlit, HuggingFace, API Development, Model Containerization & Operationalization

Core Technical Skills

Python, SQL/MySQL, Java, C/C++, HTML5, SSRS.

PROFESSIONAL EXPERIENCE

Patton Labs Inc. - Client: Oakland County, Pontiac, MI

Technical IT Business Analyst (Contractor) January 2015 – Present

- Partnered with Health and Human Services, Homeland Security, and Water Resource Commission stakeholders to architect, manage, and maintain enterprise and web applications supporting County operations.
- Interviewed business stakeholders and end users to uncover pain points, gathered requirements, and translated them into Epics, features, user stories, and acceptance criteria to drive Agile application development.
- Developed test cases and test plans for functional, integration, and User Acceptance Testing (UAT); led UAT sessions and delivered end-user training prior to project go-lives.
- Built SSRS reports to deliver data-driven insight into Health Department programs and processes, enabling evidence-based business reporting.
- Led cross-functional stakeholder and IT teams in defining and selecting product evaluation metrics to assess and implement COTS products replacing legacy applications.

AI/ML APPLIED PROJECTS & INITIATIVES

Great Learning/ University of Texas at Austin — AI/ML Program, 2025- 2026

GitHub repo - https://github.com/sgovind5/Post-Graduate-Program-AI-ML-Projects

Medical Assistant — Generative AI & RAG

- Developed a secure, production-ready **Retrieval-Augmented Generation (RAG)** prototype leveraging **Large Language Models (LLMs)** and custom **prompt engineering** to process complex medical manuals, mitigating information overload and standardizing diagnostic workflows for end users.

SuperKart Forecasting & Deployment Pipeline — MLOps

- Engineered an end-to-end **predictive ML pipeline** to forecast regional sales revenue across a multi-tier supermarket retail chain; **containerized the model using Docker** and built an interactive **Streamlit** dashboard with **Flask APIs** to operationalize data-driven insights at scale.

HelmNet Automated Compliance System — Computer Vision

- Built an automated industrial image analysis system using **Convolutional Neural Networks (CNNs)**, transfer learning, and data augmentation to detect safety helmet compliance in real time, reducing human error and automating oversight in high-risk physical environments.

ReneWind Predictive Maintenance System — Neural Networks

- Created and optimized **deep neural network classification models** using wind turbine generator sensor data to proactively predict mechanical failures, reducing false negatives to minimize operational downtime and improve cost efficiency.

EasyVisa Applicant Prediction Framework — Advanced Machine Learning

- Built and evaluated high-performance **ensemble classification models (Bagging, Boosting, Stacking)** to predict visa application approval status, generating business recommendations from high-influence attribute analysis to streamline the vetting process.

Personal Loan Campaign — Predictive Modeling

- Analyzed bank customer attributes and built a **Decision Tree classification model** to identify customers with high likelihood of purchasing a personal loan, applying **EDA**, data pre-processing, and model performance evaluation to generate business recommendations that improved marketing targeting and conversion efforts.

FoodHub — Exploratory Data Analysis

- Conducted **exploratory data analysis** using **Python, NumPy, Pandas, and Seaborn** on a food aggregator dataset, performing univariate and bivariate analysis on restaurant and cuisine demand to deliver actionable business recommendations for improving customer experience.

Dell EMC Corporation, Bangalore, India
Lead Software Developer January 2009 – December 2012

- Architected and built a tool to automate the modeling of network and storage devices within the EMC SMARTS suite, streamlining device onboarding for engineering teams.
- Built a search engine to look up MIB data, driver information, and frequent issues for devices modeled in EMC Smarts, reducing service desk support hours and support tickets to the development team by 40%.
- Created proof-of-concepts and prototypes to validate and de-risk new product features ahead of development investment.
- Built and maintained SAN root cause analysis software to monitor and identify the source and impact of availability problems across storage and network environments.